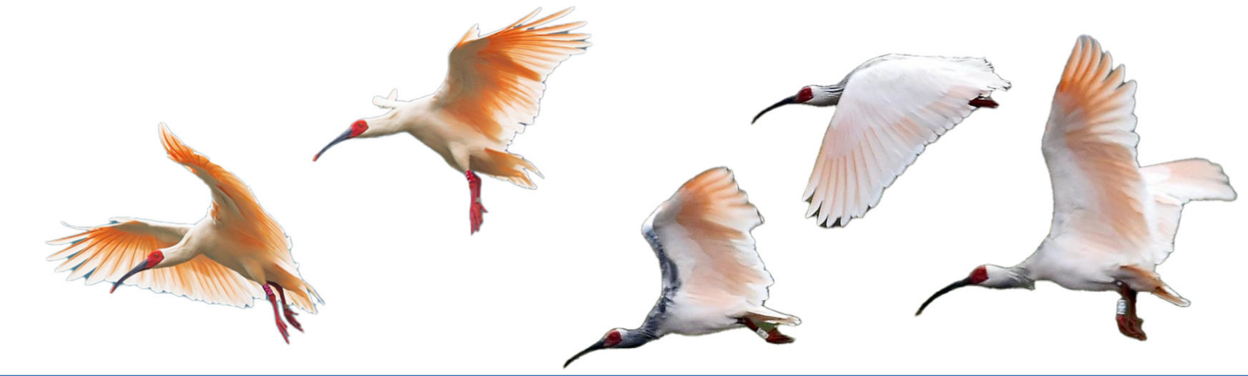
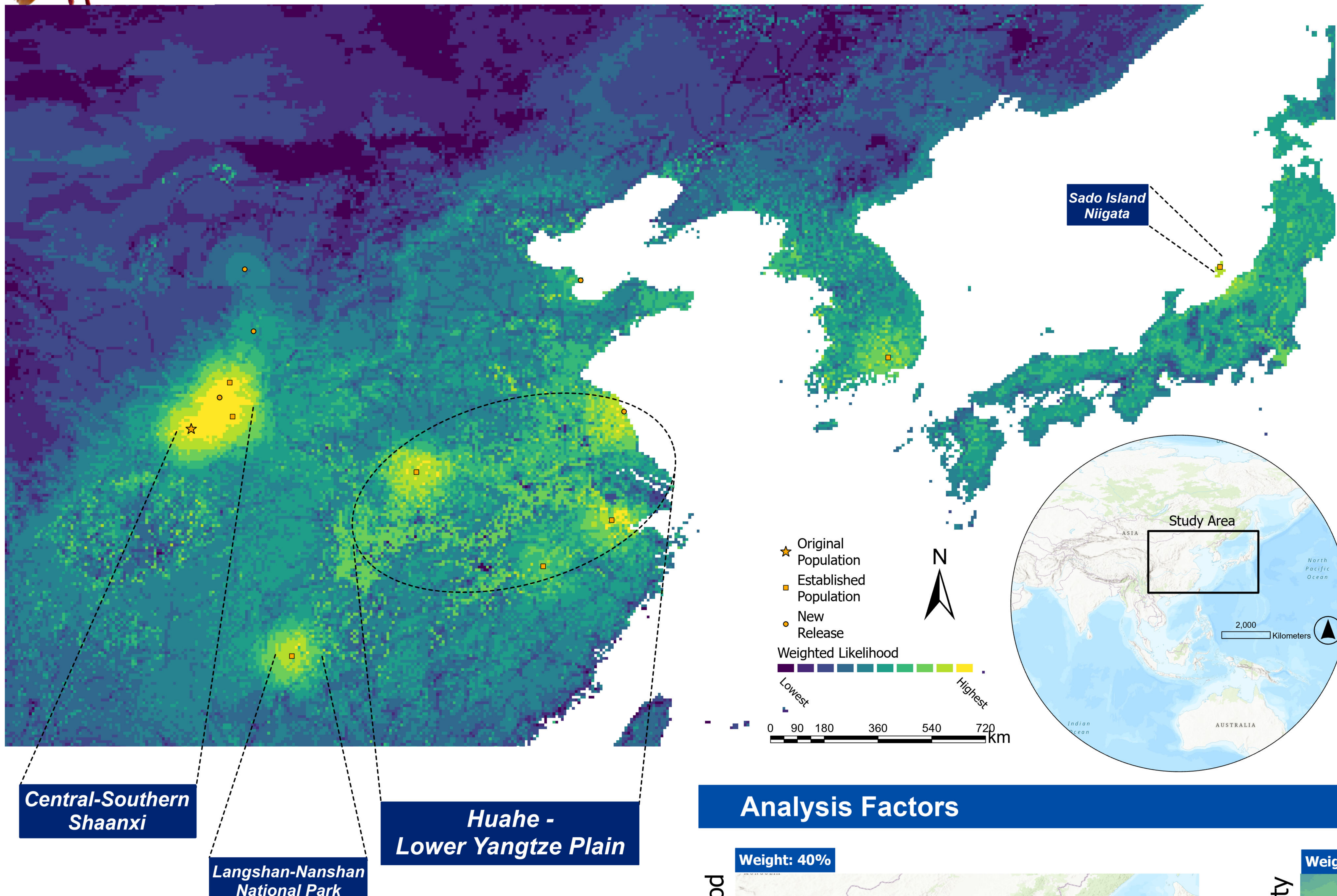
Recovering Lost Lands

Projecting Dispersal and Future Distribution Range of Crested Ibis





Background: Crested Ibis Revived from the Edge of Extinction

The crested ibis (Nipponia nippon) was once widespread across East Asia, including central and eastern China, and most areas of Korea and Japan. It depends on freshwater wetlands to feed and tall trees to roost and reproduce. Historically, as humans developed natural wetlands into paddy rice fields, crested ibis adapted to depend mainly on rice fields as their feed ground. After every fall's harvest, farmers reserved water in the paddy to make next spring's work start easier, and crested ibis relied on such winter paddies to forage. Starting from about 1950s, the industrialization of East Asian nations made most farmers have access to machines and reservoirs, and can get water in the spring easily. So, most paddies no longer reserve water during the winter. Moreover, trees were massively cut down. Consequently, crested ibis' range shrank to mountainous areas of central China, where farmlands were too fragmented to implement modern practices and winter water paddies still existed.

In the 1980s, crested ibis was thought to be extinct until a population of less than 50 individuals were found in Yangxian County, Shaanxi Province, central China. Since then, intensive conservation efforts were made and this last population has largely increased and expanded. Moreover, captive population of crested ibis were built, and captive-bred individuals were reintroduced to multiple locations in crested ibis' former range. Some captive-bred individuals were given to Korea and Japan, where the species had gone locally extinct, for reintroduction. Today, the species' population has exceeded 5000.

Crested ibis is expanding so fast that its range is already beyond the range projected by earlier studies. Given the rapid change of crested ibis' status and range, it is important to predict where will this species expand to using latest data.

Methods: Suitable Habitat + Proximity to Source Population

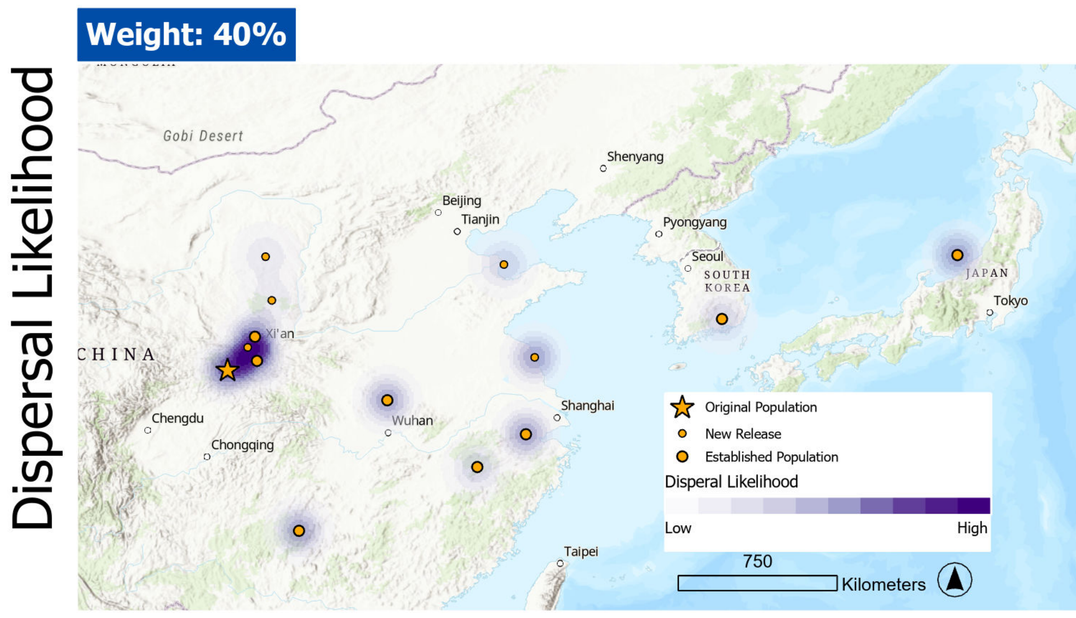
The future distribution of crested ibis is controlled by both habitat suitability and the location from which captive-bred individuals are reintroduced. Regarding habitat suitability, I read 7 papers and found that it is controlled by: availability of wetlands which provides feeding grounds, availability of tall trees to roost and reproduce, climate, elevation, and human disturbance measured by proximity to towns, cities, or major roads. Regarding source populations, I incorporated the original population in Yangxian County, 8 reintroduction sites where viable wild populations have established, and 5 reintroduction sites where individuals were newly released.

Historically, crested ibis lost much of its habitats due to advancement of farming practice. In Yangxian County, the government subsidizes farmers to farm traditionally so that crested ibises keep having water paddies in the winter. This method is to costly to be widely implemented. So, outside of Yangxian, how can reintroduced crested ibises find habitats? Luckily, successful reintroduction proved that crested ibis has regained their ancestors' capacity of living in natural wetlands. As a result, this analysis calculates the density of natural wetlands for the wetland availability part.

Analysis Factors

Dispersal Likelihood

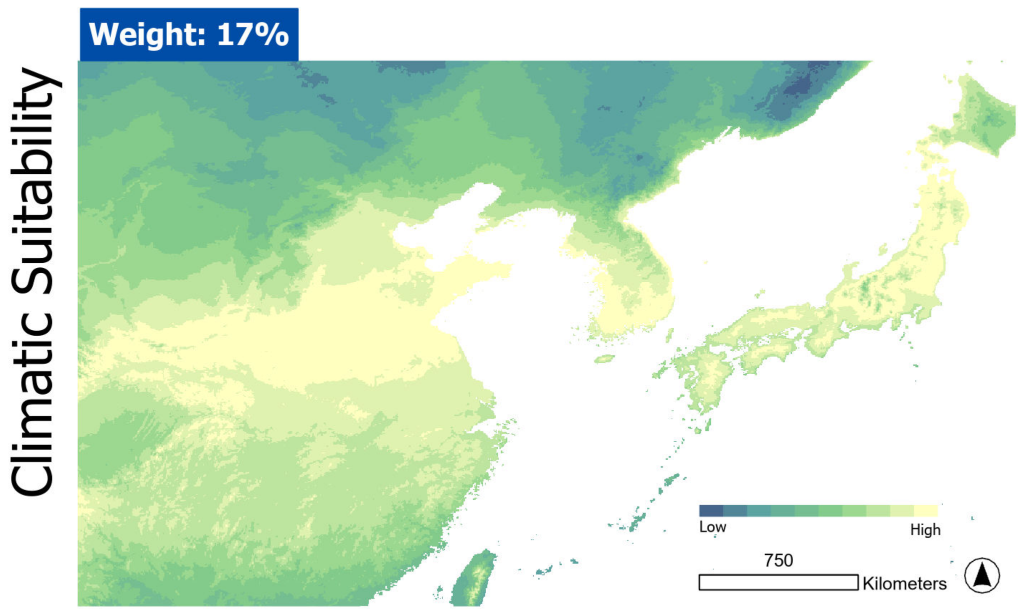
Weight: 40%



In the 1990s. there were only 1 crested ibis population existing. It is in Yanxian County, marked by the star in the map. As the population recovers, individuals disperse from this original population for new habitats, and nearby areas have a high likelihood of having dispersed crested ibises.
Captive-bred individuals are also being released at multiple locations. For locations where reintroduced individuals have established a wild population, there is also a high likelihood of dispersal in nearby areas. For locations where crest ibis is newly reintroduced, there is a potential of dispersal in the future.

Climatic Suitability

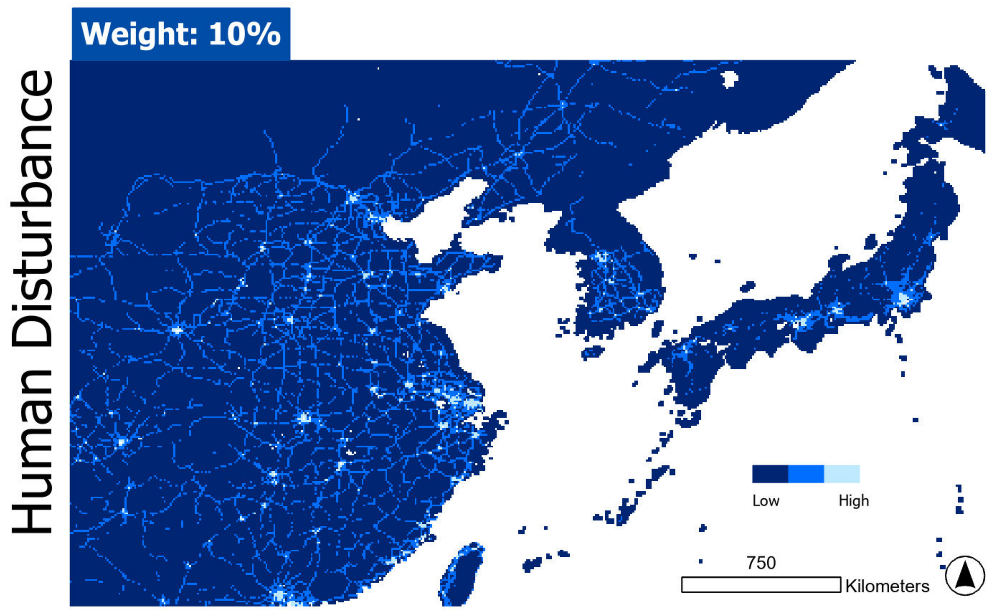
Weight: 17%



Studies have shown that the climate is one of the major factors in shaping the distribution of crested ibis. Specifically, the climate is most ideal for crested ibis when the minimum temperature of the coldest month is about -2°C and the mean temperature of the coldest season is about 0°C.
When the minimum temperature of the coldest month is -10°C or lower, wetlands freeze in the winter, making crested ibis unable to forage. Crested ibis can neither tolerate a hot climate with the minimum temperature higher than 10°C.

Human Disturbance

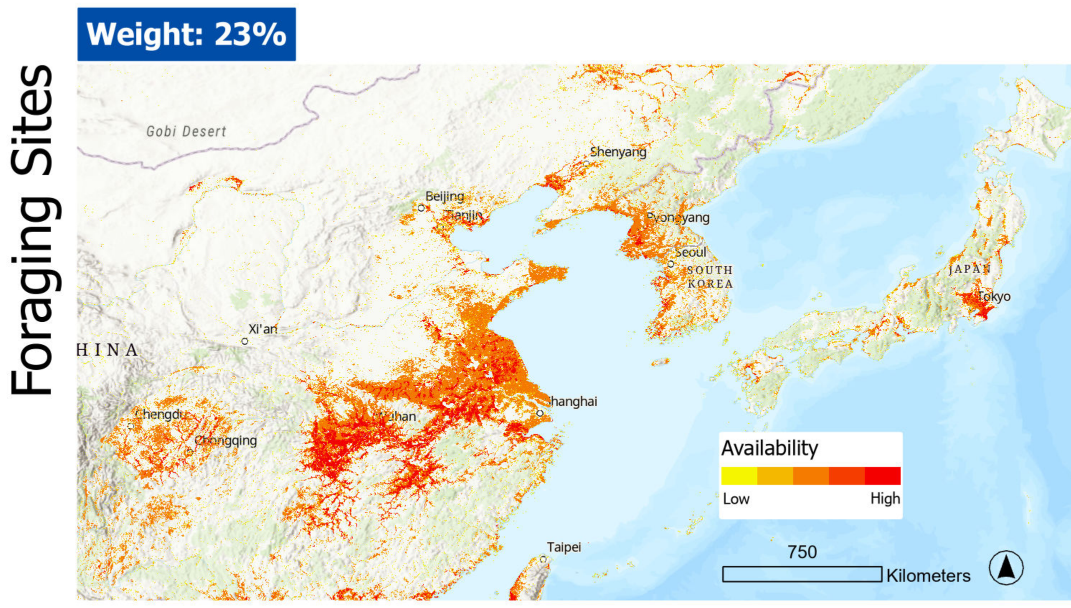
Weight: 10%



Unlike other endangered bird species, the crested ibis is tolerant of many forms of human activities. They are able to forage in paddy fields in the presence of working farmers, and can also roost in trees next to the village during the night. The form of human disturbance they cannot tolerate is large-scale architecture, including towns, cities, and major roads.
Crested ibises almost always live more than 500 meters away from major roads, and more than 1 km away from towns and cities.

Foraging Sites

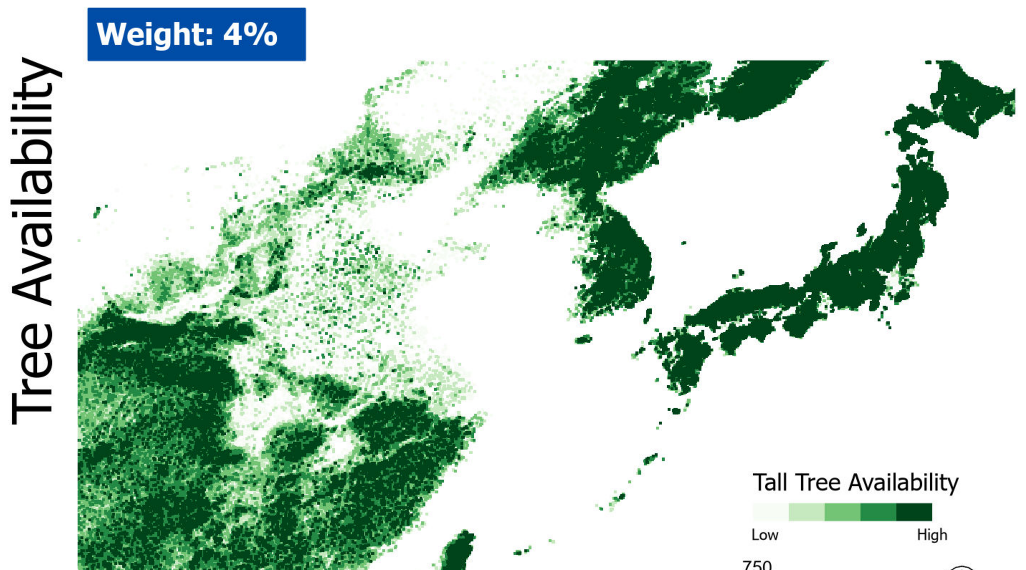
Weight: 23%



Crested ibis feed primarily on amphibians, small fish, water snails, bivalves, and other aquatic invertebrates from wetlands' shallow waters. As a result, they can only survive near wetland areas, including paddy rice fields, riverine, river delta, swamps, and wetlands around lakes or reservoirs.

Tree Availability

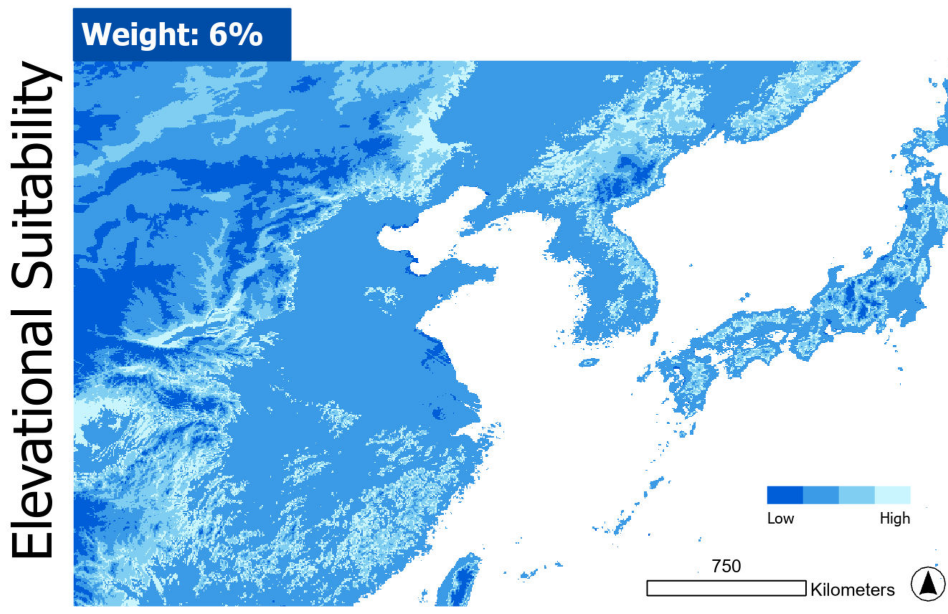
Weight: 4%



To avoid predation and disturbance, crested ibises sleep over the night on tall trees. They also make nests on tall trees to raise the young. Crested ibis nests are usually 12 meters above the ground, and the tree they nest on needs to be at least 15 meters tall.

Elevational Suitability

Weight: 6%



The only existing natural population of crested ibis lives between the elevation of 600 meters and 1400 meters above sea level. However, this might be a biased range because crested ibises lost their low-elevation habitats due to human influences. Historically, crested ibises were mainly found between 50 meters and 1000 meters above sea level. So, I set the overlap of historical range and current range as the most suitable elevation range, the range only covered by historical or current range as the medium suitable elevation range, and the range even higher or lower as not suitable.

Discussion: Where to Expect Crested Ibis in Near Future?

In the near future, crested ibis would recover from a critically endangered species whose range is smaller than a county, to a less-endangered species that occurs in many places of central and eastern China, and maybe in Korea and Japan as well.
In China's Shaanxi Province, with crested ibis's origin population and multiple successful reintroduction sites, there would be a major hotspot of crested ibis. The plain formed by Huaihe River and lower part of Yangtze River, with its extensive wetlands and suitable climate conditions, would be the major range for crested ibis to expand into. 4 reintroduction populations are starting in this area, and one day we might see these four populations' ranges connect to cover the whole plain. Another reintroduced population is starting in Langshan-Nanshan National Park. Here, crested ibis uses fragmented wetlands in the mountains. These suitable sites are not as visible as a river plain on the map, but still indicates crested ibis' potential of expanding its range into the mountainous southeastern China.
In Japan, the reintroduced population is isolated on the Sado island. However, it is reported that crested ibis has some capacity of flying over the sea, thus potentially being able to disperse to northern Honshu Island, where suitable habitats are available at multiple locations.
In Korea, the circumstance is similar to that in southeastern China: suitable wetland habitats are fragmented in the mountains, and the introduced crested ibis population in Upo Wetland, South Gyeongsang, has the potential of dispersing to these sites in the future.

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Data Sources: HydroSHEDS GLWDv2; GLAD Canopy Height 2019; WorldClim; GRIP Global Roads
Projection: WGS 1984
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